Aldus Test Document

A L E X   T I A N

> This document tests every feature Aldus must handle: headings, math, tables, images, code blocks, blockquotes, and lists.

Table of Contents

Headings

Heading 3

Heading 4

Inline Math

The quadratic formula is $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

The area of a circle is $A = \pi r^2$, and the circumference is $C = 2\pi r$.

Einstein's famous equation: $E = mc^2$.

The slope of a line: $m = \frac{y_2 - y_1}{x_2 - x_1}$.

Display Math

The Pythagorean theorem:

$$a^2 + b^2 = c^2$$

The sine law:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

The quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

A summation:

$$S_n = \sum_{i=1}^{n} ar^{i-1} = \frac{a(1 - r^n)}{1 - r}, \quad r \neq 1$$

A limit:

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

An integral:

$$\int_a^b f(x)\, dx = F(b) - F(a)$$

Tables

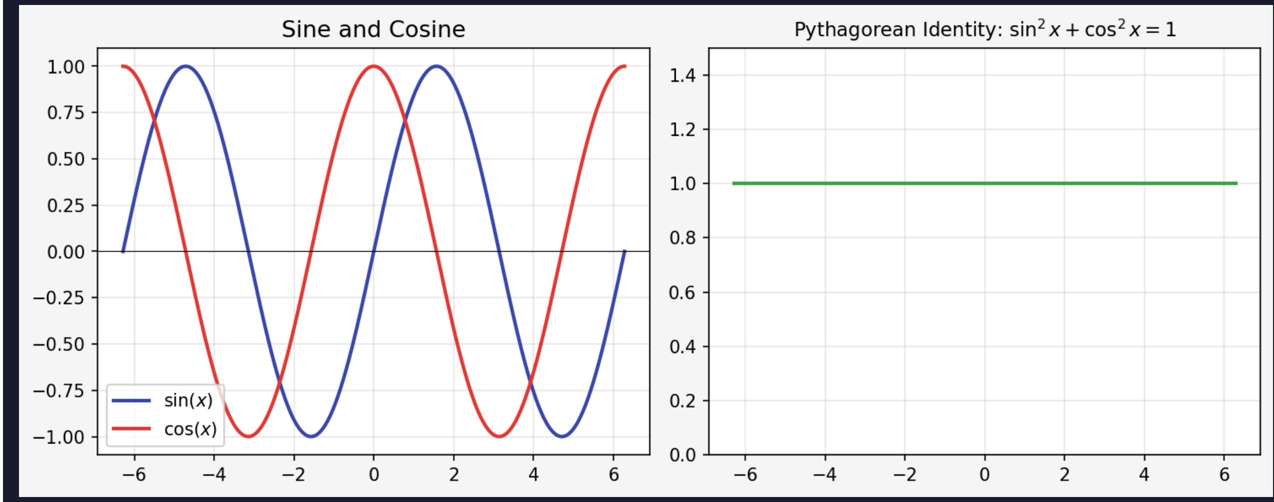
Feature	Status	Notes
LaTeX math	✅	Inline and display
Images	✅	Base64 embedded
Tables	✅	With alternating rows
Code blocks	✅	Monospace font
Blockquotes	✅	Highlighted left border

A math table:

Shape	Area Formula	Perimeter Formula
Circle	$A = \pi r^2$	$C = 2\pi r$
Rectangle	$A = lw$	$P = 2(l + w)$
Triangle	$A = \frac{1}{2}bh$	$P = a + b + c$
Sphere (surface)	$A = 4\pi r^2$	—

Images

The image below is embedded as base64 — no external file dependency in the PDF.



Code Blocks

Inline code: `f(x) = x^2 + 3x - 4`

A Python code block:

```
def quadratic(a, b, c):
    discriminant = b**2 - 4*a*c
    if discriminant < 0:
        return None
    x1 = (-b + discriminant**0.5) / (2*a)
    x2 = (-b - discriminant**0.5) / (2*a)
    return x1, x2
```

A plain text block:

```
Step 1: Identify a, b, c from ax² + bx + c = 0
Step 2: Calculate the discriminant b² - 4ac
Step 3: If discriminant < 0, no real solutions
Step 4: Apply the quadratic formula
```

Blockquotes

> **Key Insight:** The discriminant $b^2 - 4ac$ tells you how many real solutions a quadratic equation has.
>
> **Note:** If $b^2 - 4ac > 0$, there are two real solutions. If $b^2 - 4ac = 0$, there is exactly one. If $b^2 - 4ac < 0$, there are no real solutions.

Lists

Unordered

- First item
- Second item with **bold text**
- Third item with $inline math: x^2$
- Fourth item with `inline code`

Ordered

1. Identify the type of problem
2. List all known variables
3. Choose the correct formula
4. Solve algebraically
5. Verify your answer

Nested

- Kinematics
- Displacement: $\Delta x = x_f - x_i$
- Velocity: $v = \dfrac{\Delta x}{\Delta t}$
- Acceleration: $a = \dfrac{\Delta v}{\Delta t}$
- Dynamics
- Newton's Second Law: $F = ma$
- Weight: $W = mg$

Text Formatting

This is **bold text**, this is *italic text*, and this is ***bold italic***.

This is ~~strikethrough~~ text.

Mixed Math and Text

To solve $2x^2 - 4x - 6 = 0$, first divide by 2:

$$x^2 - 2x - 3 = 0$$

Factor the left side:

$$(x-3)(x+1) = 0$$

Therefore $x = 3$ or $x = -1$.

We can verify using the quadratic formula with $a = 1$, $b = -2$, $c = -3$:

$$x = \frac{2 \pm \sqrt{4+12}}{2} = \frac{2 \pm 4}{2}$$

Which gives $x = 3$ or $x = -1$. ✓

End of test document.
